PROFESSIONAL PORTFOLIO PROJECT

HR Analytics Dashboard

Employee Attrition Analysis | Microsoft Power BI

TOOL	DATASET	LAYOUT	AUDIENCE
Power BI Desktop	CSV · 1,480 records	Single-page, interactive	HR team & professionals

01 PROJECT OVERVIEW

An interactive Power BI dashboard that analyzes employee attrition, salary distribution, job satisfaction, and workforce demographics to give HR teams a clear, data-driven view of turnover.

02 PROJECT OBJECTIVE

Identify workforce trends and potential factors associated with employee turnover, supporting data-driven HR decisions and employee retention.

03 DATASET DESCRIPTION

The CSV dataset contains 1,480 employee records covering demographics, department, job role, salary, work experience, years at the company, job satisfaction, performance, overtime, and attrition status, as applicable to the available fields.

04 TOOLS AND TECHNOLOGIES

Technology	Purpose
Power BI Desktop	Dashboard development, KPI cards, and data visualization
Power Query	Data cleaning, transformation, and preparation
DAX	Measures for employee attrition and workforce analysis
CSV	Source dataset

05 DATA CLEANING AND TRANSFORMATION

- **Duplicates:** removed duplicate records to improve data consistency.
- **Data types:** corrected types to support accurate calculations and analysis.
- **Column names:** renamed columns for readability and consistency.
- **Derived fields:** created conditional or calculated columns for additional insights.
- **Other:** performed further transformations to prepare the data for dashboard development.

06 DASHBOARD DESIGN

Visualizations

- **Bar charts** to compare attrition across categories.
- **Column charts** to analyze workforce metrics.
- **Pie / donut charts** to present categorical distributions.
- **Stacked bar / column charts** to compare workforce segments.

Interactivity

- **Age slicer** for age-based workforce analysis.
- **Department slicer** to filter employee data by department.
- **Cross-filtering** so one selection updates related visuals.
- **Dynamic KPIs and tooltips** that respond to filters and hover.

07 DASHBOARD KPIS

TOTAL EMPLOYEES	ATTRITION COUNT	ATTRITION RATE	AVERAGE SALARY
1,417	231	16.30%	[Not on dashboard]
AVERAGE AGE	**AVG. EXPERIENCE (YEARS)**	**HIGHEST-ATTRITION DEPT.**	**ACTIVE EMPLOYEES**
37	7.04	Administration	1,186

KPI values taken from the dashboard (Figure 1). Average Salary is not shown on the dashboard and remains to be verified.

08 DAX MEASURES

- **Attrition Count:** number of employees who left the organization.
- **Attrition Rate:** proportion of employees who left, using the defined DAX logic.
- **Other measures:** [Verify names and formulas].

Formulas intentionally omitted until confirmed from the Power BI model.

09 KEY ANALYSIS AREAS

- **Department & job role:** attrition differences across departments and roles.
- **Salary:** pay differences between employees who left and who stayed.
- **Experience & tenure:** attrition by years of experience and time at the company.
- **Demographics:** age-based filtering of workforce composition and attrition.

10 KEY FINDINGS

- **Department:** Administration has the most attrition (84 of 231, 36.36%), followed by Sales (52, 22.51%) and Operations (43, 18.61%).
- **Job role:** Laboratory Technician (59) and Sales Executive (56) have the highest attrition counts, followed by Research Scientist (45).
- **Salary slab:** the 6-10 LPA band has the highest attrition count; the 10+ LPA band has the lowest.
- **Gender and tenure:** male employees account for 146 leavers (63.2%) and female for 85 (36.8%); attrition peaks at low years of experience.

11 RECOMMENDATIONS

- Investigate departments and job roles with comparatively higher attrition, using verified dashboard results.
- Review salary patterns, career development, and engagement where the data indicates retention risk.
- Use workforce metrics to support targeted retention strategies, and monitor attrition trends for workforce planning.

12 CONCLUSION

The dashboard turns employee data into interactive insights, helping HR professionals investigate attrition patterns, assess workforce trends, and make informed decisions to improve retention and organizational performance.

13 DASHBOARD SCREENSHOT



Figure 1: HR Analytics Dashboard – Employee Attrition Analysis.